

Deployment

Deployment Guide & Verification

This document specifies the deployment sequence, environment variables, deployment scripts, supported networks, verification processes, and post-deployment validation steps for UnifyVault V2.

1. Supported Networks

- **Network Name:** Chain ID | RPC URL Environment Variable | Block Explorer
- ****Base Mainnet**:** `8453` | `BASE\_MAINNET\_RPC\_URL` | <https://basescan.org>
- ****Base Sepolia**:** `84532` | `BASE\_SEPOLIA\_RPC\_URL` | <https://sepolia.basescan.org>

2. Deployment Prerequisites & Environment Variables

The following environment variables must be exported prior to running deployment scripts:

```
# RPC Node Endpoints
BASE_SEPOLIA_RPC_URL=https://sepolia.base.org
BASE_MAINNET_RPC_URL=https://mainnet.base.org

# Deployer & Admin Accounts
PRIVATE_KEY=0x... # Deployer private key
NEXT_PUBLIC_ADMIN_ADDRESS=0xd905920c91853039060246Ed5724AA72B91a96DA
GNOSIS_SAFE_ADDRESS=0x1111111111111111111111111111111111111111111111111111111111111111 # Multi-sig proposal manager for Timelock

# Etherscan / Basescan Verification API Key
BASESCAN_API_KEY=your_basescan_api_key
```

3. Mandatory Deployment Sequence

```
flowchart TD
    Step1[1. Deploy ProtocolDirectory] --> Step2[2. Deploy OracleManager & ChainlinkOracleProvider]
    Step2 --> Step3[3. Deploy Treasury & FeeManager]
    Step3 --> Step4[4. Deploy CustodyVault & LiquidityManager]
    Step4 --> Step5[5. Deploy CostBasisManagerV2 & PerformanceManager]
    Step5 --> Step6[6. Deploy UVBEV2 Token & Attach CostBasisManager Hook]
    Step6 --> Step7[7. Deploy UnifyVaultTimelock 48h Delay]
    Step7 --> Step8[8. Deploy StrategyManager, PortfolioManager, & SwapAdapter]
    Step8 --> Step9[9. Deploy UnifyVaultController]
    Step9 --> Step10[10. Deploy P2PEscrowV2 & Marketplace]
    Step10 --> Step11[11. Register Modules in ProtocolDirectory & Set Escrow Status]
    Step11 --> Step12[12. Grant Controller Roles & Transfer Admin to Timelock]
```

4. Deployment Scripts

Primary deployment scripts are located in `packages/protocol/script`:

- ****`DeployV2Production.s.sol`**:** Complete V2 production deployment including `CostBasisManagerV2`, `UVBEV2`, `P2PEscrowV2`, and `Marketplace`.
- ****`DeployMainnet.s.sol`**:** Production deployment script for Base Mainnet utilizing real Chainlink oracle feeds and DEX routers.

- **ReconcileBaseSepoliaState.s.sol**: State reconciliation and configuration script for Base Sepolia testnet.
 - **mainnet/GrantAdminRoles.s.sol**: Assigns `GOVERNANCE_ROLE` and `DEFAULT_ADMIN_ROLE` to `UnifyVaultTimelock`.
 - **mainnet/RenounceOldAdmin.s.sol**: Revokes deployer permissions after governance transfer.
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5. Execution Commands

5.1 Local Simulation (No On-Chain State Change)

```
cd packages/protocol
forge script script/DeployV2Production.s.sol:DeployV2ProductionScript
```

5.2 Testnet Deployment (Base Sepolia)

```
cd packages/protocol
forge script script/DeployV2Production.s.sol:DeployV2ProductionScript \
  --rpc-url $BASE_SEPOLIA_RPC_URL \
  --private-key $PRIVATE_KEY \
  --broadcast \
  --verify \
  --etherscan-api-key $BASESCAN_API_KEY
```

5.3 Production Mainnet Deployment (Base Mainnet)

```
cd packages/protocol
forge script script/DeployMainnet.s.sol:DeployMainnetScript \
  --rpc-url $BASE_MAINNET_RPC_URL \
  --private-key $PRIVATE_KEY \
  --broadcast \
  --verify \
  --etherscan-api-key $BASESCAN_API_KEY
```

6. Canonical Deployment Addresses (Base Sepolia Testnet - Chain ID 84532)

- **Contract Module**: Canonical Address
 - **ProtocolDirectory**: `0xD2715141a0F5998B707BaA963990bFC2E94cF145``
 - **Treasury**: `0x66182F56BD5E523c655f6890290aB519f528e83f``
 - **FeeManager**: `0x0721465B01b586B7AAAdF957A4a884acE46CfbEc9``
 - **CustodyVault**: `0x27B5C6DEA90678B78856b0B10DBA37A789fDe97e``
 - **OracleManager**: `0x5B6067982C6ccE2DC760EB4731c1b40136776D4A``
 - **ChainlinkOracleProvider**: `0x4F7f99653d9d7aCD462429ffFc0C4B6C8Cf4354a``
 - **LiquidityManager**: `0xa938aaCeA64bE8f41c90960aFF232dA4Df7Fc329``
 - **UVBEV2 (UVBEToken)**: `0xA3Db7c3DeE9A50D966A06e19b5DF4FCDee615BdE``
 - **UnifyVaultController**: `0x07f3D3432B64DBF67c5b061AF2bC8Aef70221Cea``
 - **StrategyManager**: `0x14058459198a2CfFc8cE89C364334a80Da82D6a3``
 - **PortfolioManager**: `0x1C65B1667c8cC03138b8e57cDd40b0Bf28a4cDc4``
 - **SwapAdapter**: `0xCb1a434c5ebe2F2F8672Ca507Ee819C6888ae634``
 - **CostBasisManagerV2**: `0xF71706A2Fd8692e3C739855B2A33C0E679b4c382``
 - **PerformanceManager**: `0x133fD024EA635694A223e66B936c2afAB4F2DB78``
 - **P2PEscrowV2**: `0xbAc9C1b440adf74688abBD5be950ABd2766E5B7b``
 - **TimelockController**: `0x9094145Cd2AEA2f309eDf14237444a07edF98d02``
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7. Post-Deployment Checklist

- [x] Every module address registered in `ProtocolDirectory`.
- [x] Oracles configured with active feeds and non-zero heartbeat limits.
- [x] `CONTROLLER_ROLE` granted to `UnifyVaultController` on `CustodyVault`, `Treasury`, `UVBEV2`, and `CostBasisManagerV2`.

- [x] `CostBasisManagerV2.setEscrowStatus(P2PEscrowAddress, true)` executed to isolate P2P transfers.
- [x] Initial deposit executed to mint and burn `DEAD_SHARES` (`1000` wei).
- [x] Frontend environment variables updated with canonical contract addresses.